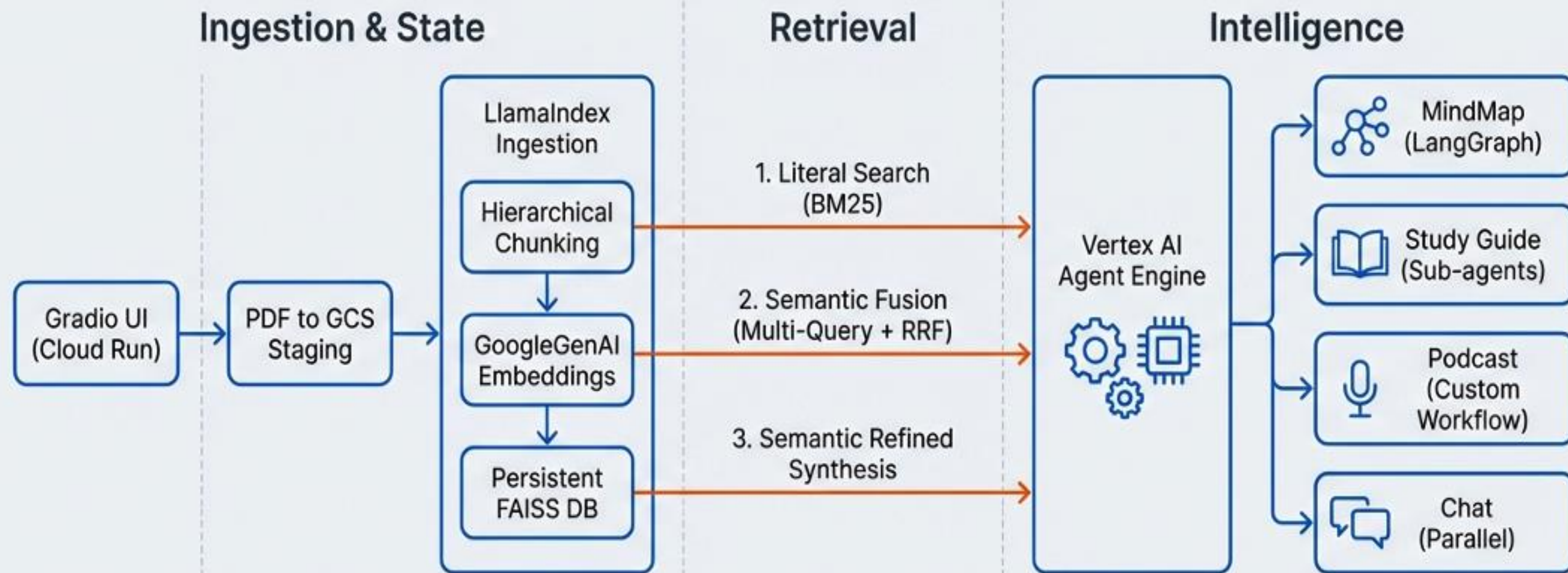


Evolution: From Database Lineage to Agentic Orchestration



I do not just prompt models; I treat GenAI applications as enterprise-grade, 3-tier systems. Wielding large-scale relational databases provided the deep data lineage expertise now used for complex RAG vector optimizations.

Distributed RAG Architecture & Multi-Agent Research Orchestrator



Tech Stack

Gradio

Cloud Run

GCS

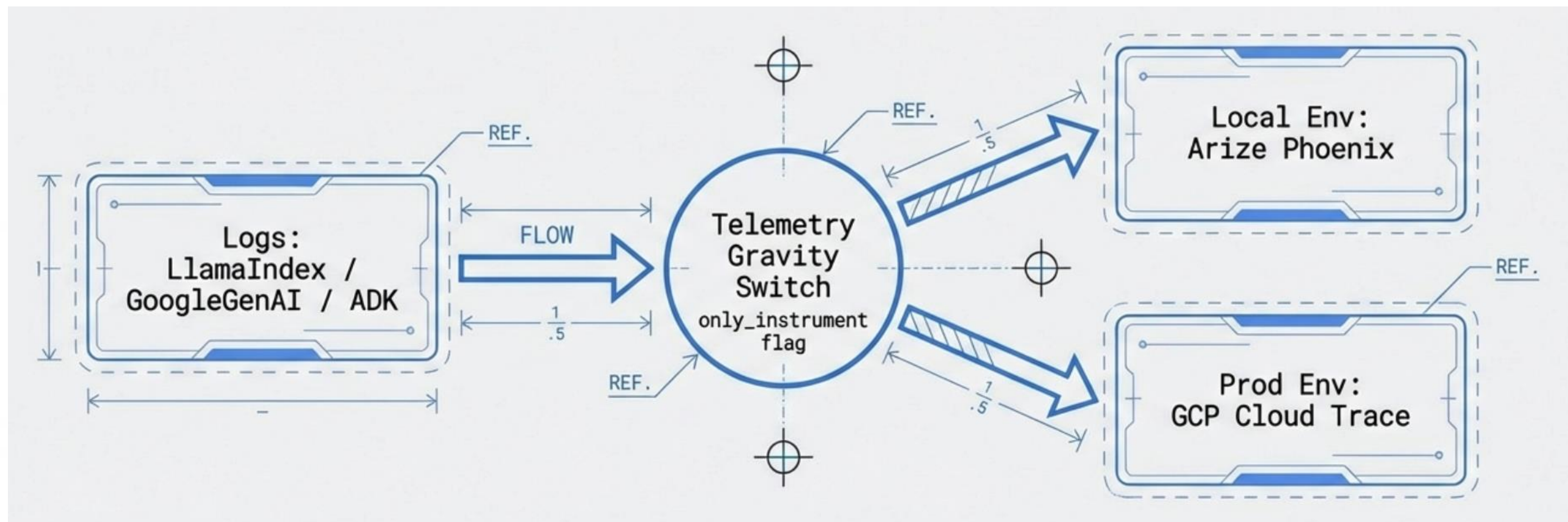
Vertex AI

LlamaIndex

FAISS

LangGraph

Telemetry Gravity: Unified Observability Pipeline



Abstracted instrumentation using an 'only_instrument' flag. The exact same code pipes granular spans to different destinations based on the environment.

Quality Assurance: High-Fidelity Custom Evaluations



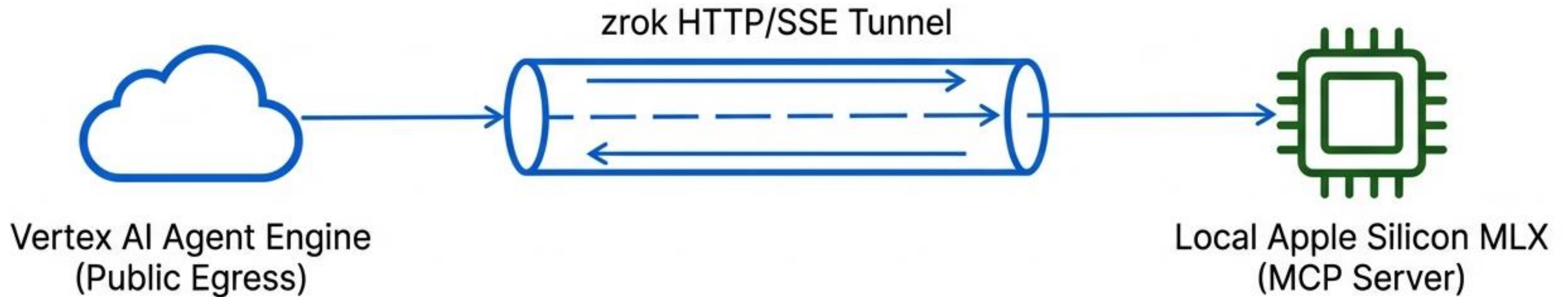
Standard ADK Registry Limitation.
(Built-in evaluator crashed on
Custom Workflow Agents).



Custom JSON Rubric Evaluator.

Engineered an “LLM-as-a-Judge” pipeline against manual JSON rubrics. Replaced 287 lines of broken framework code with a 10-line Python iterator to validate tool execution sequences.

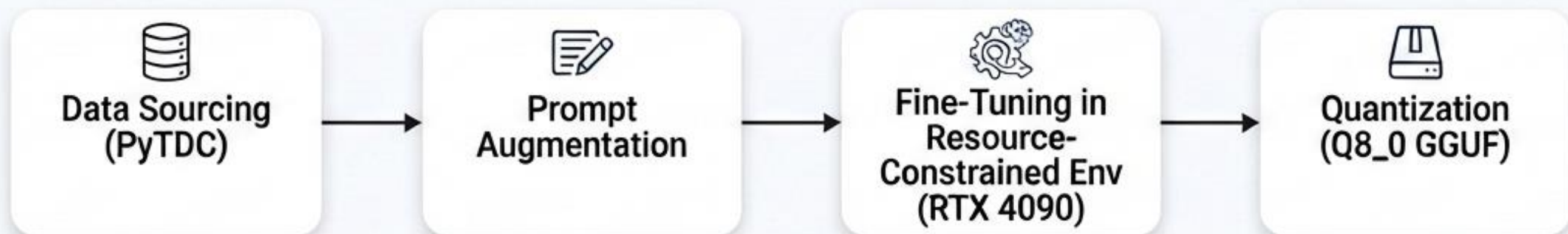
Hybrid Networking: The Cloud-to-Edge Bridge



- Context: Vertex AI Linux VMs cannot compile Apple instructions required for the Podcast agent's audio synthesis.
- Solution: Migrated local MCP server to use Server-Sent Events (SSE) and exposed via a stable zrok tunnel, routing cloud reasoning to isolated local hardware.

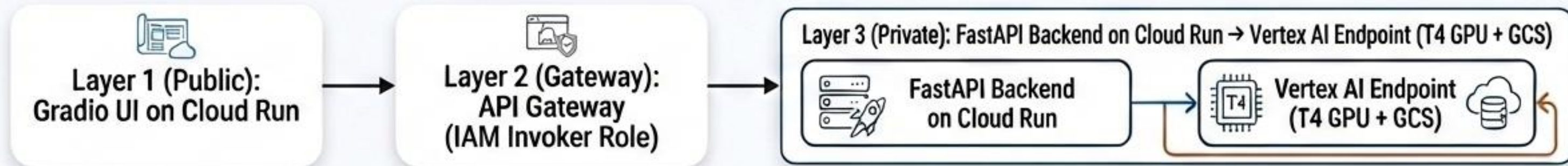
Specialized Drug Interaction LLM (Fine-Tuning & Serving)

Part 1: Fine-Tuning Pipeline



331.68% Performance Increase (Score: 8.245 vs Base Model: 1.91).

Part 2: 3-Tier Enterprise Deployment (Serving)

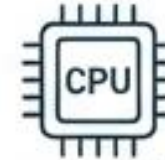


Backend securely queries Vertex Endpoint running custom 'llama-cpp-python' container.
Container auto-provisions by downloading GGUF model from GCS on boot.

Impact Analysis & Return on Investment

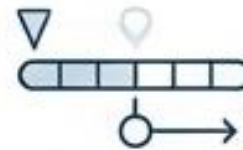
331.68%

Performance Increase
(DDI Fine-Tuning)



~2 Min

Generation Time
(15-Min Podcast)



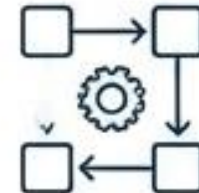
92%

Pass Rate
(Custom Faithfulness Evals)



80%

Runner Complexity Reduction
(MCP Smart Tools)



**Architecting systems that deliver measurable ROI and survive enterprise reality.
Structurally sound, financially efficient, and relentlessly secure.**

ROI

Performance

Efficiency

Security